

Evaluation Problem 3

Group C21

Duration: 2 hrs

You are designing a memory management unit (MMU). The system has a fixed number of Physical Frames (e.g., 3 or 4). A process generates a Reference String (a sequence of integers representing page requests). Your job is to determine which pages are in memory at any given time and count the total number of Page Faults.

Logical Clock: Maintain a global integer counter that starts at 0. Every time a page is accessed (whether it's a hit or a fault), increment the counter.

Page Hit: If the requested page is already in a frame, update that frame's "Last Used Time" to the current counter.

Page Fault (Empty Frame): If the page is not in memory and there is an empty frame (indicated by -1), load the page and set its "Last Used Time" to the current counter.

Page Fault (Eviction): If memory is full, find the frame that has the smallest (oldest) "Last Used Time." Replace that page with the new one and update its "Last Used Time" to the current counter.